



Whitepaper

The AI-Native Financial Institution

How AWS and Databricks
Power Intelligence at Scale



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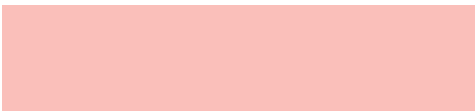
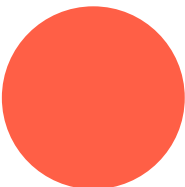
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Just a few years ago, most financial services institutions (FSI) ran AI as a series of contained experiments: proofs of concept that were funded out of innovation budgets and scoped to a single team, rarely connected to production systems. That posture is no longer tenable. According to “[Making AI Deliver](#),” an Economist Enterprise report commissioned by Databricks, approximately three in five leading AI adopters across industries now have autonomous agents doing real work. However, more than one in five financial services firms have not begun exploring AI agents at all.

AI AGENT ADOPTION	% OF ADOPTION
Leading AI Adopters Accross Industries Autonomous agents doing real work	60%
Financial Services Firms An opportunity for AI implementation	20%

Source: Economist Enterprise, cited in Making AI deliver (2026)

The result is a widening gap between what the technology can do and what regulated institutions are positioned to capture. Banks, insurers, asset managers and payments firms are racing to close it by operationalizing intelligence across the enterprise.

Several forces are accelerating this change. Fintechs continue to raise customer expectations around speed, personalization and self-service. Regulators are demanding clear documentation of how models are trained, how decisions are reached and how outcomes are monitored in production. At the same time, persistent margin pressure is pushing every line of business to do more with less, making intelligent automation less of an aspiration and more of a requirement.

The bottlenecks have also shifted. Frontier models are widely accessible, and their capabilities improve on a near-monthly cadence. The challenge in moving AI into production now sits earlier in the process: data fragmented across legacy systems, governance applied unevenly across teams and infrastructure that strains when a pilot scales across an enterprise.

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Today, FSI investment in AI is centered around revenue growth, operational efficiency and a stronger risk and compliance posture. Each one depends on the same underlying conditions: trusted data, governed access and infrastructure that scales securely under regulatory load. Firms that unify those layers compound their advantage with every new use case they bring online, from fraud detection to underwriting to client-facing assistants. Firms that are still navigating fragmented data and ad hoc controls fall further behind with every cycle.

The institutions that are leading the AI race aren't the ones with the flashiest models. They're the ones with the data foundations, governance frameworks and infrastructure that enable AI to run reliably at scale.

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Together, AWS and Databricks support a unified approach to building an AI stack for financial services. AWS delivers secure, scalable infrastructure and Databricks provides the intelligence layer that makes that infrastructure actionable. They form a vertically integrated stack in which every layer, from storage and compute to models and business interfaces, works as a unified system.

At the infrastructure layer, AWS and Databricks operate as a pair. Databricks workloads benefit from the durable, scalable object storage and the elastic compute that Amazon S3 and Amazon EC2 respectively provide. Workloads stay efficient and economical with AWS Graviton delivering price-performance gains, and AWS Trainium accelerates the model training and fine-tuning that Databricks Agent Bricks orchestrates.

A mobile financial services platform with 57 million monthly active users runs on this foundation. Using Databricks with Delta Lake running natively on AWS, the financial services platform built a more flexible and consistent data infrastructure for its short-term financing data. This improved pipeline stability, supporting real-time data processing and sharpening decision-making across the organization, all while managing costs as data volumes grow.

Governance is built into the stack at every level, not bolted on afterward. AWS IAM, Amazon VPC, and end-to-end encryption form the security perimeter. Inside that perimeter, Unity Catalog extends governance across every data asset, model, and AI workload, so that the controls AWS establishes at the infrastructure level are reinforced and made visible at the data and AI level by Databricks. The documentation, lineage and access records that regulators expect are generated as a byproduct of how the platform operates, rather than as a separate compliance effort.

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Model access is where flexibility meets control. Amazon Bedrock provides governed, single-interface access to a curated set of foundation models, including Anthropic's Claude, so teams can evaluate, switch, and deploy the best model for each use case without re-engineering integrations. Databricks Agent Bricks and MLflow then take over: handling training, fine-tuning, live deployment, and ongoing performance monitoring inside the same governed environment that AWS secures. Nothing falls outside the controls that protect the rest of the platform.

The top of the stack is where AI reaches the business. Databricks Genie provides nontechnical users with a natural language interface to ask questions and get answers in seconds. A relationship manager, risk officer or product lead can pull the insight that they need without waiting on a data or AI team because the underlying data, governed by Unity Catalog and secured by AWS, is always audit ready, current, and trusted.

Databricks Apps and Jobs, together with the AWS partner community, embed AI-powered insights and actions directly into the front-office tools where decisions actually get made, whether it's the underwriting platform, the trading desk or the customer service console.

While AWS provides the scalable, secure infrastructure foundation, Databricks is what makes that infrastructure intelligent. It governs every data asset, operationalizes every model and delivers answers to every business user through a single, unified environment.

**Amazon Bedrock****MODEL ACCESS**

Evaluate • Compare •
Select Models

**databricks****MODEL DEVELOPMENT**

Train • Fine-Tune • Experiment

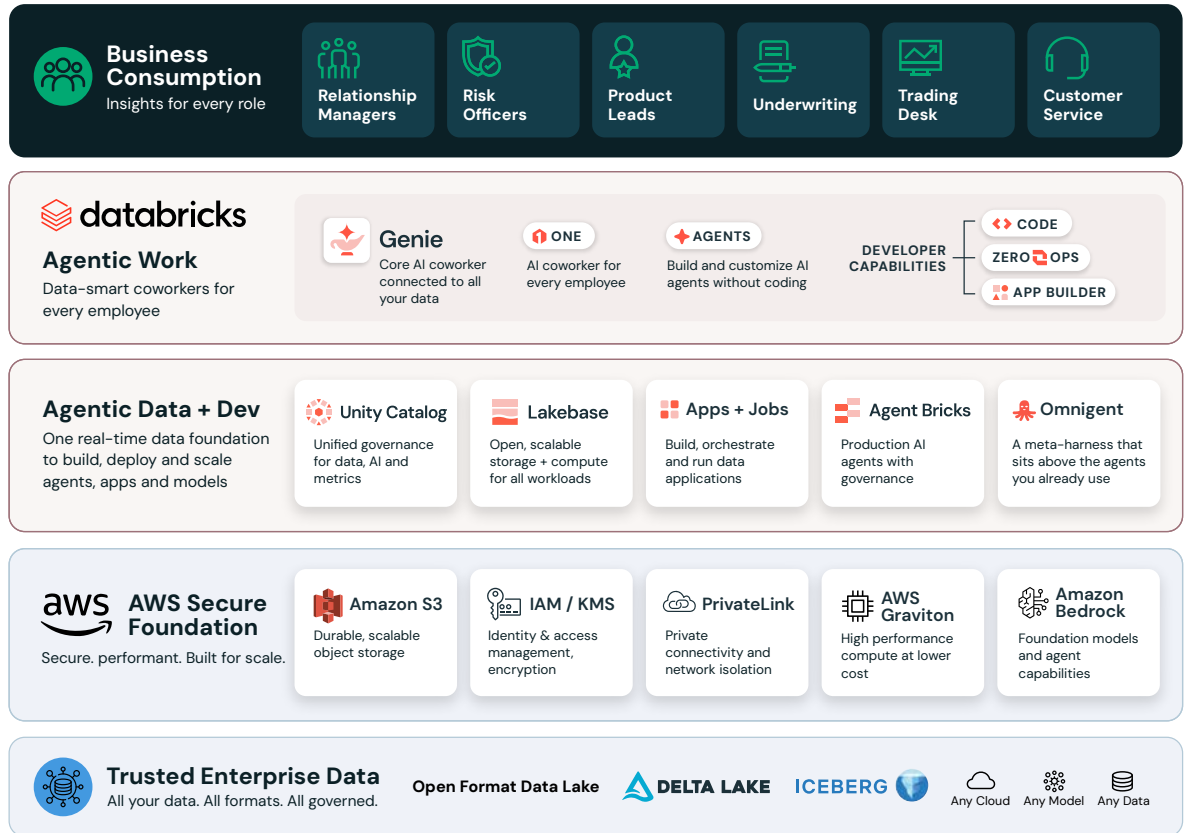
**MODEL OPERATIONS**

Deploy • Monitor • Optimize

**GOVERNANCE LAYER**

Security • Compliance •
Controls

A unified AI stack built for FSI — Context. Control. Cost. Choice.



Four differentiators set this combined stack apart

- **Running Databricks on AWS Graviton** delivers measurable price-performance gains, lowering TCO without compromising the scale FSIs need.
- **Unity Catalog and Amazon S3 together** create a single governed pipeline. Raw data in AWS becomes a queryable, lineage-tracked asset in Databricks with no re-governance step in between.
- **Databricks Agent Bricks turns model access through Amazon Bedrock** into production-grade AI, with full lifecycle management inside the same governed environment.
- **Databricks in AWS Marketplace** simplifies procurement: one vendor relationship, one cloud spend line, drawing on existing AWS commitments with no new contracts required.

Every layer of this AI stack reinforces the next. Data flows from Amazon S3 into Databricks Lakehouse without losing governance. Models trained or accessed through Amazon Bedrock inherit the same controls. Business users query the same governed assets through Genie. And the entire system runs on infrastructure tuned for the cost, performance and security profile that FSIs require.

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On its own, AWS infrastructure offers powerful compute and scalability, but organizations require an additional layer to unify data access and model governance. The Databricks platform without AWS loses the elastic, cost-efficient infrastructure that makes enterprise-scale AI economical. The value of the combined stack is that neither constraint applies.

The AWS-Databricks stack is already powering production AI across every major FSI sub-vertical, and the use cases below show how the same foundation adapts to each workload's requirements.

Real-time fraud detection | Payments

Fraud detection requires FSIs to process massive transaction volumes, score each one in milliseconds and get the answer right the first time. Transaction data flows from Amazon S3 into Spark™ Declarative Pipelines (SDP), where it's enriched, scored and routed in near real time. Models hosted through Amazon Bedrock flag transactions that deviate from established patterns, while Unity Catalog logs every decision and traces it back to the data and model version that produced it.

Coinbase relies on machine learning to detect fraud and mitigate money laundering risk across its cryptocurrency exchange platform, where milliseconds can affect both security and trust. Running on Databricks, the company moved its critical risk models from microbatch streaming to Spark Real-Time Mode, cutting latency to well under 100ms and improving feature consistency by up to 98%. Faster, more consistent fraud decisions are now delivered at scale, with compute costs cut in half.

SITUATION	APPROACH	IMPACT
Real-time fraud and anti-money-laundering risk detection requires subsecond latency and high accuracy	Migrated risk models from microbatch streaming to Spark Real-Time Mode on Databricks	<100ms latency, 98% better feature consistency and compute costs cut in half

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AI-assisted underwriting | Insurance

Underwriting has historically been bottlenecked by the difficulty of unifying structured and unstructured data, such as claims histories, third-party data, application documents and external risk signals, that is scattered across separate systems. Unity Catalog brings these sources into a single governed environment where Anthropic’s Claude, accessed through Amazon Bedrock, scores risk against the full picture rather than a fragment of it.

The outcome is faster, more consistent underwriting decisions, with the lineage and audit trail that regulators require built into every output. Underwriters spend less time assembling context and more time exercising judgment on the cases that need it.

SITUATION	APPROACH	IMPACT
Fragmented underwriting inputs	Unified governed environment	Better underwriting with faster decisions, consistency, built-in audit trails and human judgment focused on complex cases

Continuous risk monitoring | Capital markets

Capital markets firms need to assess portfolio risk continuously across multiple asset classes, with results required in seconds. A model pipeline orchestrated in Lakeflow Jobs runs on AWS infrastructure tuned for latency-sensitive compute. Real-time market data streams into the lakehouse, risk models score positions and exposures and results surface to traders and risk officers without the overnight batch lag that legacy systems experience.

The same governance layer that supports compliance reporting also supports faster decision-making at the desk. Risk teams can query positions, scenarios and exposures through the same governed interface that auditors use, with no separate environment or duplicate dataset to maintain.

SITUATION	APPROACH	IMPACT
Delayed overnight risk processing + fragmented systems	Real-time governed risk pipeline on AWS infrastructure	Faster decisions, unified governance, and real-time exposure visibility

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Natural-language market insights | Asset management and trading

Most enterprise data is still locked behind SQL. Analysts who can write queries get answers in minutes, while everyone else waits days for a data and AI team to fulfill their requests. Genie removes that bottleneck by giving business users a natural-language interface to query governed enterprise data.

A portfolio manager can ask about exposure to a specific sector, a trader can pull historical performance under comparable market conditions and an operations lead can check fund-level metrics; all in plain language, all governed by Unity Catalog and all without writing a line of SQL. Behind the scenes, Genie taps into Anthropic’s Claude to interpret questions and translate them into governed queries against the underlying data.

Addepar, a global technology and data platform for wealth and investment management, models more than \$8 trillion in client investments and portfolios globally. Running Databricks on AWS, Addepar has achieved \$2 million in infrastructure cost savings and a 5x improvement in the ability to deliver new pipelines and integrations. Unity Catalog and shared notebooks give Addepar’s teams a single place to work across data, code and SQL.



SITUATION	APPROACH	IMPACT
\$8T+ in modeled investments Global portfolio operations Complex data workflows	Unified data platform on AWS with shared notebooks, and a single workspace for data, code and SQL	\$2M infrastructure savings 5x faster pipeline delivery Improved team collaboration

Regulatory reporting and compliance automation | Cross-vertical

According to Deloitte’s 2025 banking and capital markets outlook, regulatory compliance costs are one of the core drivers of swelling expense budgets at major US banks and are unlikely to decline anytime soon. Regulatory reporting returns no direct revenue and expands every time a new rule takes effect, but the combined stack streamlines it in two ways:

- Unity Catalog tracks lineage end to end. Every figure in every report is traceable back through the queries, models and datasets that produced it, giving examiners the audit trail that they expect.
- LLM-assisted tooling generates first-draft reports, narratives and disclosures from that governed data, reducing the manual lift on compliance teams.

FSIs gain shorter reporting cycles, fewer manual reconciliation steps and faster, evidence-backed responses to regulator queries.

Financial technology company nCino has built a platform that’s trusted by more than 2,800 customers with trillions of dollars in processed loan history. Running Databricks on AWS, nCino keeps all banking data within a single controlled environment, reducing compliance risk and lowering the complexity of stitching together multiple systems. Unity Catalog enforces guardrails and governance across the platform, while Genie democratizes data access; enabling executives, analysts, relationship managers and service teams to tap into nCino’s deep loan lifecycle history without technical expertise.

		
SITUATION	APPROACH	IMPACT
2,800+ customers with trillions in loan history and requires strict governance with fragmented financial systems	Single governed banking data environment with platform guardrails + governed access, and a shared interface for business teams	Reduced compliance risk with lower system complexity, democratized data access, and faster trusted decisions

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Databricks and AWS have made joint investments aimed at moving workloads from architecture diagrams to production systems.

Strategic partnership at scale

Databricks passed \$2 billion in total contract value in **AWS Marketplace** during the 2025 calendar year, with Marketplace business growing roughly 95% year over year. For FSIs evaluating procurement paths, this scale corresponds to consolidated billing, faster contracting and a single vendor relationship covering both the cloud infrastructure and the data intelligence platform running on it. In 2025, Databricks received seven AWS Partner of the Year awards, reflecting the depth of the technical and go-to-market relationship.

SCALE

\$2B TCV + 95% growth

CUSTOMER VALUE

simplified procurement +
vendor consolidation

CREDIBILITY

7 awards + strategic
partnership depth

OPTIMIZATION

Reduced compute costs+
accelerated model fine-tuning

UNIFIED FOUNDATION

Optimized enterprise data
movement into analytical and
AI workloads

ENTERPRISE OUTCOMES

Improved performance, lower
operational complexity, and
greater scalability

LONG-TERM COMMITMENT

Strategic alignment, shared
innovation, and FSI investment

Engineering AI infrastructure for enterprise performance

Joint engineering investments are aimed at making the combined stack faster, more cost-efficient and easier to operate. Running Databricks on AWS Graviton lowers the cost of running compute workloads and AWS Trainium helps accelerate training when teams fine-tune their own models. For enterprise-scale data movement into Databricks Lakebase, a managed database layer for modern analytical and AI workloads, the Amazon S3 connector is now generally available on AWS. The 2024 AWS Strategic Collaboration Agreement set out the partnership's go-to-market commitment, with dedicated joint teams, shared solution blueprints and coordinated investment in the FSI vertical.



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One governed environment for enterprise AI

AWS and Databricks maintain dedicated go-to-market teams, joint presence at industry events and shared blueprints for key use cases.

Foundation models, including Anthropic's Claude, can be accessed through Amazon Bedrock, so FSIs can evaluate, compare and deploy different models for each workload, within the same governed environment used for other assets on the platform.

For FSIs, an integrated foundation backed by joint investment contributes to a lower integration risk, faster time-to-value and a clearer path from pilot to production than a stack assembled from unrelated vendors typically allows.

Databricks contributes the intelligence layer that makes those infrastructure investments actionable: the governance, the model operations and the business-facing interfaces that work alongside AWS infrastructure to support production AI.

STRATEGIC COLLABORATION

Joint teams + shared AI blueprints

MODEL FLEXIBILITY

Compare and deploy multiple foundation models

GOVERNED PLATFORM

One secure environment across AI and data assets

FSI VALUE

Faster adoption and lower operational complexity

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The FSIs pulling ahead are the ones treating AI as core operating infrastructure rather than a series of isolated projects. Databricks on AWS provides the foundation to scale AI across the enterprise with governed data, flexible model access and scalable infrastructure, all designed to work together from the first deployment to the hundredth.

The next step is choosing where to start.

Talk to your representative today about how you
can run AI reliably at scale with Databricks and AWS.

To learn more about Databricks and AWS in FSI, visit

[**Databricks for Financial Services and AWS for Financial Services**](#)